

I-system field recordings

this piece can be tested by arranging pink noise

- each iteration can have more high end freqs cut to distinguish individual voices more

field recordings to be arranged

- divide into groups
 - grass/field/meadow
 - leaves/trees
 - water
 - birds/wildlife?
- use these groups to determine a theme or themes of a generated version of the piece

arrangement

- mix and match by ratio/proportion
 - e.g. 60% woods, 25% meadow, 15% river
- some options creating environments that are entirely meadow or woodland
 - **could also correspond with visuals - select footage of the same environments with the same proportions?**
- try to arrange spatially - possibly using a particular branch of the I-system
 - therefore one particular branch of meadows may be panned mostly left and one of rivers centre-right, both surrounded by woodlands
- **angle** of lines of the I-system going left and right can correspond to left and right **panning** of sounds
 - as time progresses and a point along a line progresses left or right, the panning moves in that direction - meaning the panning is constantly shifting to the left or right - as a moving cross section of the line represented by a point would be
- the **number of iterations** corresponds to a **decrease in volume**
 - as lines get shorter and more numerous with iteration of the I-system's rules the volumes of the corresponding recordings being played decreases
 - this complements that their panning may also be more extreme
- **line length** also corresponds to the **length in time of a field recording segment**
 - the first field recording selected will likely be the longest, and segments of others will be derived from the length of the longest one
 - all lengths could be standardised both in the I-system and field recordings/recording segments
- smaller offshoots towards the ends of the branches can be small snippets of the same recording as the line they branched off from
 - parent child branch relationship past a certain number of iterations? (also determine using chance - increasing as iteration increases?)

how it might be generated

- the full l-system rendering may need to be generated before the sound begins
 - this will allow the full proportions and randomised elements to be ready for use
- before introducing randomness it can be drafted using a simpler symmetrical version of the l-system:

$X \rightarrow F[+X][-X]FX$

$F \rightarrow FF$

- in this l-system instead of F meaning draw forward (draw a line), it can mean to increase duration (by _ seconds?)
- every new set of square brackets [] will be the introduction of a new sample - the type of sample (category: meadow, woods, water, etc) can be determined by chance (based on proportions to represent a determined environment (from a set of fixed proportions?))
 - after a certain number of iterations/nested brackets - new samples may stop being added, and instead smaller segments of the parent branch's sample may be used - would be introduced using conditionals (perhaps introduced by changing probability?)
- each + and -, or other changing of angle determines panning (within the limit of furthest pan possible/desirable)
 - given both duration of a branch and angle - the pan can move over time rather than be immediately fixed to the furthest pan of a particular branch
- with number of iterations and duration volume can also be decided
- duration can be relative to the length of the first field recording, which will be the length of the full piece and longest and central "line" of the l-system
 - these first recordings can be called "stem" recordings
 - the first recording will last the length of the whole piece
 - stem recordings must be over a certain length
- all other durations will be derived from the length of the stem recording, bc they will all be proportional to each other
- the l-system will have to be generated as a usable series of numbers representing each branch: duration/length, panning/angle, iteration
- do in supercollider?